

TechFacts

#103

Switching of transformers with vacuum circuit-breakers

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Blue technology
with Zero F-gases

1 Introduction

Circuit-breakers are widely used for energizing and de-energizing transformers. A distinction can be made between operation under normal system conditions and operation under fault conditions. Under normal operation, the following switching operations must be ensured:

- De-energization of unloaded transformers,
- De-energization of reactor-loaded transformers.

The breaking of transformer-limited short-circuits is of interest for operation under fault conditions. In high-voltage systems, SF₆ circuit-breakers are the dominant technology. The behavior of these circuit-breakers and their interaction with transformers are well-studied and understood. The limitations of the SF₆ technology are well-known as well. Hereafter, the suitability of vacuum circuit-breakers for the above-mentioned switching duties shall be illustrated. The limitations are discussed, and recommendations are given.

2 Operation under normal conditions

2.1 De-energization of unloaded transformers

The de-energization of unloaded transformers is an insignificant duty for a circuit-breaker [1]. The magnetizing current to be interrupted is at most in the range of few amperes.

Type tests are not foreseen for the switching of transformers due to the following reasons [2]:

- The non-linear behavior of a transformer core cannot be simulated by linear elements in a test laboratory,
- Tests on a certain transformer are not representative of other transformers,
- Transformer switching is not as demanding as other inductive load switching duties.

When a transformer (represented as an RLC circuit) is de-energized, the surge capacitance remains charged and decays to zero as an underdamped oscillation. The amplitude factor does not usually exceed 1.3 p.u. and the frequency is in the range of a few hundred hertz [1]; see Figure 1.

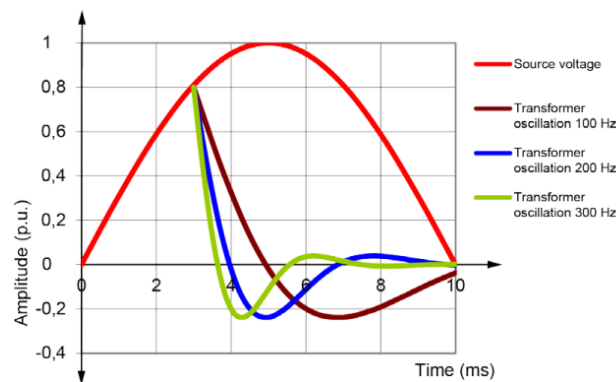


Figure 1: Transformer oscillation [1].

The transient recovery voltage (TRV) appearing across the circuit-breaker is the difference between the transformer voltage and the system voltage, see Figure 2. The simplicity of the operation becomes obvious when compared to the de-energization of shunt reactors: their

natural frequency is in the kilohertz range and the amplitude factor goes up to 1.9 p.u. for air-core reactors.

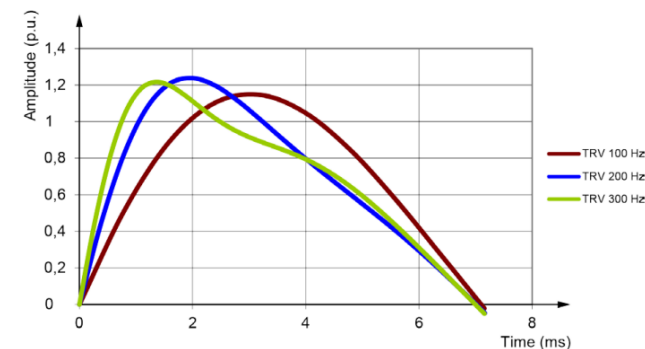


Figure 2: TRV across circuit-breaker [1].

The applicability of vacuum circuit-breakers for such duties is demonstrated in technical papers based on system measurements, since there is a lack of applicable type tests.

Comparison measurements on vacuum, SF₆, and compressed air circuit-breakers have been performed and evaluated [3].

Figure 3 shows the load side voltages of a 110 kV/10.5 kV transformer during de-energization with an SF₆ circuit-breaker. The behavior is very similar to what is shown in Figure 1. The transformer voltage decays exponentially, and the amplitude factor is far less than 1.3 p.u.

In the blue phase two re-ignitions occurred. If more than one re-ignition occurs, we call it multiple re-ignitions. This behavior is well-known and does not pose any problem, neither to the circuit-breaker nor to the transformer or power system. Even though these re-ignitions occur, there is no evidence of any remarkable overvoltage.

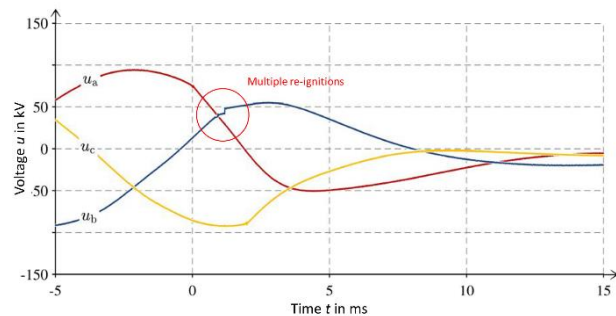


Figure 3: Transformer voltages during de-energization with an SF₆ circuit-breaker [3].

Figure 4 shows the load side voltages of a 110 kV/31.5 kV transformer during de-energization with a vacuum circuit-breaker.

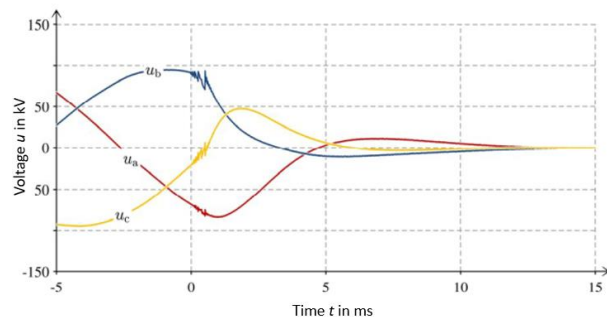


Figure 4: Transformer voltages during de-energization with a vacuum circuit-breaker [3].

As already found during de-energization with the SF₆ circuit-breaker, few re-ignitions (multiple or single) occurred. Even though with the vacuum circuit-breaker the number of re-ignitions is slightly higher, there were no significant overvoltages detected.

Due to losses in the transformer (iron core and eddy currents), and the fact that the overall stored energy in the transformer is low, the probability of considerable overvoltages is low as well [4].

2.2 De-energization of reactor-loaded transformers

Two relevant cases can be considered for the switching of reactor-loaded transformers.

The first one is the reactor-loaded tertiary winding of a network interconnecting transformer. The tertiary winding of such transformers is often used to provide the reactive power of the transmission lines connected to the primary and/or secondary side of the transformer. If the primary or the secondary side of such a transformer were to be disconnected, the remaining circuit-breaker on the secondary or the primary side would have to switch a reactor-loaded transformer.

The likelihood of such a configuration is rather low, since the system interconnecting transformers must provide the system interconnection and maintain power transfer.

For the occasional case in which this configuration might occur, it can be considered as uncritical. The number of re-ignitions and the overvoltage that might occur are effectively limited due to the rather high capacitance of the transformer and the losses in the core.

Additional mitigation measures are required in the case where a two-winding transformer, switched on the primary side, is only dedicated to providing reactive power and is to be operated frequently. This case should be treated as shunt reactor switching, and RC filter circuits must be installed between the transformer and circuit-

breaker. As an alternative, point-on-wave switching may be applied in the future.

The switching of shunt reactors in the presence of RC circuits was subjected to type tests and is a proven solution when operated with vacuum circuit-breakers.

3 Operation under fault conditions

3.1 General

Transformers can be affected by fault conditions in a variety of ways. If the fault current seen by the circuit-breaker interrupting the fault does not flow through the transformer winding(s), it can be considered as a terminal fault and no further considerations are required.

If the windings of one or more phases are involved in the fault, the situation is known as a transformer-limited fault. This type of fault will be discussed in the following sub-section.

There may be internal faults of transformers where only part of one or more windings is involved. This situation is something in between the above-mentioned cases and is covered either by the terminal fault or the transformer-limited fault considerations.

Transformer-limited faults are always associated with rather low fault currents because the transformer reactance has a fault-limiting effect.

3.2 Transformer-limited fault

Transformer-limited faults can be categorized into transformer-fed faults and transformer secondary faults [1], see Figures 5 and 6.

In most cases, the transformer-limited fault is covered by the terminal fault test duties T10 and T30 [5].

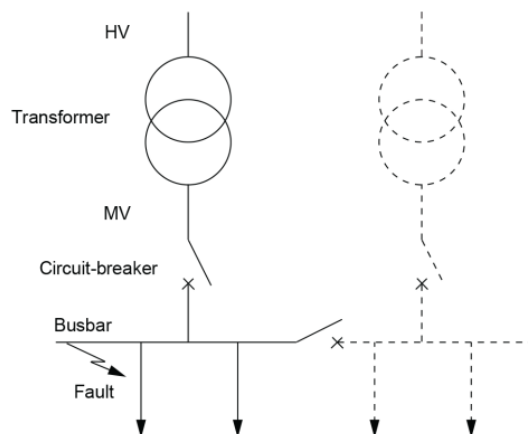


Figure 5: Example of transformer-fed fault.

In rare cases, when the capacitance of the transformer is low and there is no remarkable capacitance between the circuit-breaker and the transformer, the dominant natural frequency of the transformer can cause transient recovery voltages (TRV).

These deliver a higher rate-of-rise of recovery voltage (RRRV), with greater amplitude factors than specified for terminal fault test duties T10 and T30.

In such cases, IEEE Std C37.06.1 [6] and IEC 62271-100, Annex F [5] can be applied.

Due to the low fault current level, even modern SF₆ circuit-breaker can approach their physical limits when stressed with such steep TRV's.

In contrast, vacuum circuit-breakers take advantage of the rapid recovery of the dielectric strength after negligible heat input on the contact surface.

Recent type tests have shown that RRRVs of 12 kV/μs and more can be withstood by a single 145 kV vacuum gap. Therefore, transformer-limited faults do not pose any challenge to high-voltage vacuum circuit-breakers.

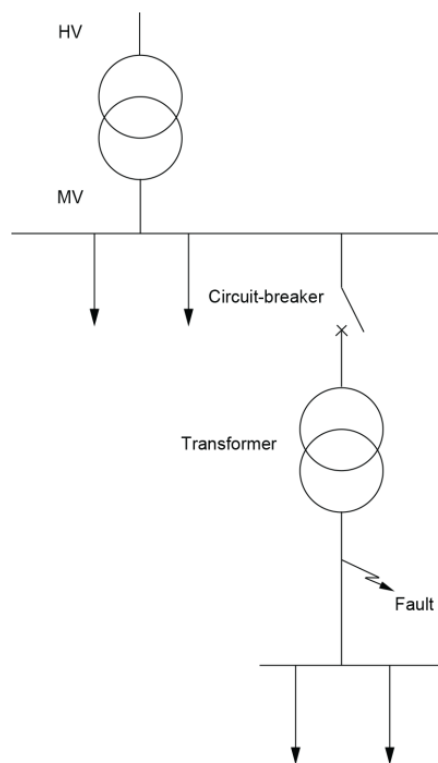


Figure 6: Example of transformer secondary fault.

4 Conclusion and recommendations

The theoretical considerations in combination with the available system measurements clearly show that high voltage vacuum circuit-breakers can be employed to switch unloaded transformers without any concern or doubt.

For reactor-loaded transformers, mitigation measures may be necessary depending on the application; but the same is also applicable to SF₆ circuit-breakers. Therefore, for such applications, the vacuum circuit-breaker offers the same advantages as the SF₆ technology.

In the case of transformer-limited faults, vacuum circuit-breakers can even take advantage of the independence between the amount of fault current, and a sufficient voltage withstand capability. Here, the vacuum technology is superior to SF₆.

It can be concluded that vacuum circuit-breakers are suitable to substitute SF₆ or any other type of circuit-breaker technology when transformers are to be switched. The statement is valid for undisturbed and disturbed (fault) system conditions.

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